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* Introduction

* Characteristics of living things

* Classification

* Reproduction

CHARACTERISTICS OF LIVING THINGS

There are 7 main activities which make an organism different from non-living things - These 7 are:

1- Nutrition or feeding: Living things take in material from their surrounding for the purpose of growth and energy supply. Nutrition can be defined as the process in which organism obtain energy and raw material from food substances such as protein, fats & oil, etc. Generally speaking, animals are heterotrophic meaning that they depend on already manufactured food from plants for their survival. Plants on the other hand are able to self-feed themselves using available chemicals namely CO_2 (carbon dioxide), sunlight and O_2 .

2- Respiration: This can be defined as the process of release of energy from food substances in all living cells. It can also be a process by which living organism obtain oxygen from the atmosphere for the purpose of breaking down food within their cells. Cellular respiration is the complete oxidation of organic compound such as glucose to extract energy from chemical bonds.

$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$$

Respiration can be aerobic or anaerobic. In animals, most organisms cannot respire without oxygen but some organisms and tissues can continue to respire if the oxygen runs out.

amphibians - mammals (in term of locomotory organelles)
are called Tetrapods

especially in the animal muscles. In animal muscles, respiration can be anaerobic for a short time. During this process, the glucose in the muscles is converted to lactic acid and energy is released in the process.

Aerobic respiration on the other hand takes place in the presence of oxygen supply.

3- Movement: All living things must move. This is the ability to move the whole or part of the body from one place to another. Different animals have different locomotive organelles for example in lower animals like protozoan, pseudopodia, cilia, flagella, in advance animal especially the vertebrates, locomotory organelles can also vary.

4- Excretion - All living things must excrete.

This is because many chemical reactions are taking place in the cell and the waste products from these chemical reactions must be removed to avoid poisoning the body. Excretion therefore can be defined as the process through which the body gets rid of metabolic wastes and maintains osmotic pressure. In majority of vertebrates, the kidney is the ~~chief~~ ^{chief} excretory organ.

g-Growth: This is a permanent increase in cell numbers and size. It involves using nutrient to produce new cells as seen in all living things

b-Reproduction: This is the biological process by which new organisms are created from their parents. The two main types are Asexual and Sexual reproduction.

The Asexual Reproduction involves a single organism reproducing without the involvement of another organism. Hence, the new organism that is produced is genetically similar to the parent

The Sexual Reproduction involves a male and female organism who are members of the same species and they combine their genetic information to create a new organism that is genetically unique. In human, reproduction involves:

i - Gamete production {sperm - male, egg - female}

ii - fertilization {male + female ^{nuclei} gamete ~~zygote~~}

fertilization is the fusion of the male and female nuclei

iii - Zygote formation: This is the process by which the fertilized egg develop into an embryo and then the fetus

1- Sensitivity / Irritability - All living things are able to sense and respond to stimuli around them, stimuli such as light, temp, gravity and chemical substances